

EXECUTIVE SUMMARY

Solving Inventory Inefficiency for Urban Retail Co.

Purpose

This project set out to improve how inventory decisions are made at Urban Retail Co. by using SQL-driven insights to surface where and why inefficiencies occur. Rather than relying on reactive restocking or generic thresholds, the aim was to build a more responsive system—one that adapts to category-level demand patterns, accounts for forecasting gaps, and highlights both stock imbalances and missed sales opportunities.

Approach

- **Structured the raw dataset** into a fully normalized relational schema (fact and dimension tables) for efficient querying and extensibility.
- **Built SQL-based pipelines** to analyze inventory behavior across dimensions like region, store, category, and product ID.
- **Defined dynamic reorder logic** using average demand × lead time, and customized thresholds for specific categories (e.g., 1.5× forecast for Home Essentials).
- **Calculated inventory turnover** and classified products into quartiles to flag low-performing SKUs and optimize space allocation.
- **Modelled stockout rates** using percentile-based inventory rankings, enabling a data-backed proxy for missed sales risks.
- **Incorporated external demand drivers** such as promotions and weather to assess their influence on forecasting accuracy.

Key Insights

Stockout & Reordering

- Nearly **22% of fast-moving electronics** in the West region faced stockouts on over a quarter of observed days.
- Many SKUs in **Home Essentials** consistently fell below their dynamic reorder threshold (calculated using 7-day forecast-based logic).

Overstock & Turnover

- Personal Care items showed low turnover (<1.0), pointing to overstocking and slow movement.
- Grocery and seasonal products performed best on turnover metrics, suggesting better alignment between stocking and demand.

Forecast Accuracy

- Forecasted sales deviated sharply from actuals during promotions, especially for electronics and seasonal items.
- Accuracy dropped significantly during **rainy and festive seasons**, likely due to unmodeled external variables.

Dynamic Reorder Logic

- Introducing category-specific reorder points (e.g., 35 units for groceries, 10 for electronics) helped reduce false low-stock flags.
- Adjustments like setting reorder thresholds to **1.5× average demand** for Home Essentials improved early warnings for critical SKUs.

Recommendations

1. **Adopt Forecast-Based Reordering**
Reorder logic should be driven by rolling demand forecasts, customized by product category.
2. **Prioritize High-Turnover SKUs**
Allocate shelf space and warehouse bandwidth to categories with faster turnover cycles.
3. **Incorporate External Demand Drivers**
Update forecasting models to include variables like seasonality, local weather, and ongoing promotions.
4. **Flag Low-Performing Inventory**
Review slow-moving SKUs (especially in Personal Care) for price correction, bundling, or removal.

Expected Business Impact

- ✓ If implemented, these insights can help Urban Retail Co.:
 - Reduce revenue loss from stockouts by 10–15%.
 - Lower inventory holding costs from excess stock.
 - Improve availability of high-demand products.
 - Strengthen supply chain agility and customer experience.

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